MILITARY SPECIFICATION

GENERATOR, DIGITAL DATA SG-1139( )/G

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND DESCRIPTION

1.1 SCOPE. This specification covers Generator, Digital Data SG-1139( )/G which is portable and provides a test capability for digital transmission systems and equipments under conditions of tactical deployment.

1.2 DESCRIPTION. The SG-1139( )/G is a piece of field test equipment used to test digital transmission paths and equipments. It performs its test by sending a repetitive test signal $2^{15} - 1$ bits long into the transmission equipment and receiving the signal after it has been over the path. It then counts the number of received bits that differ from the transmitted bits. One SG-1139( )/G can test a looped back system and two SG-1139( )/G's can test a system end-to-end.

It shall generate the digital test signal at bit rates which match the bit rates of ATACS channels (600 to 32,000 b/s), ATACS groups (576 Kb/s), TRI-TAC channels (16 and 32 Kb/s) and TRI-TAC groups (128 to 4608 Kb/s). It shall provide a balanced conditioned diphase interface to match the ATACS and TRI-TAC channels, an unbalanced NRZ interface to match the ATACS groups, and a balanced NRZ interface to match the TRI-TAC groups.

In the loopback and station clock modes and with the NRZ balanced interface, the SG-1139( )/G shall operate at any rate, externally supplied, between 2 Kb/s and 4915.2 Kb/s.

The results of the measurement shall be displayed in two ways. The first is a simple display consisting of three colored lamps. The green lamp means that the transmission is good, and it shall light if the Bit Error Rate (BER) is less than 10^{-5}. The red lamp means that the transmission is bad, and it shall light if the BER is more than 10^{-3}. The yellow lamp shall indicate a marginal transmission, that is, a BER between 10^{-5} and 10^{-3}. The second shall be a 7-digit numerical display of cumulative errors. Each error shall advance the count by one until the display reset button is pressed or the counter overflows. This shall give useful information in cases where the error rate is very low.

The SG-1139( )/G shall have internal self-checking features. BITE sensors with LED indicators shall be used to locate faulty circuit card assemblies. A single error insertion button shall be provided to insure that the complete unit is working properly.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command, ATTN: DRSEL — ED-TO, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2 DESCRIPTION (Cont.)

The SG-1139( )/G shall be designed to be used in shelters and to be transported between shelters by truck. Alternatively, it can be permanently mounted in a shelter.

The SG-1139( )/G shall be mounted in a rugged case with transit cover. Four (4) plug-in circuit card assemblies shall contain the logic and interface circuitry. Two (2) additional plug-in circuit card assemblies shall provide the required power supply. The controls, indicators, and connectors shall be on the front face of the unit, except for the power connector which shall be on the rear. Access to the system under test shall be made by connecting the appropriate connectors on the SG-1139( )/G front panel to the shelter patch field. The channel rate interfaces shall connect to the equipment under test by use of field wire. A ground stud is also provided on the rear surface of the unit.

2. APPLICABLE DOCUMENTS

2.1 ISSUES OF DOCUMENTS. The following documents form a part of this specification to the extent specified herein. The effective issue or revision of specifications and standards shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (a) specific issues are set forth herein, or (b) different issues than those specified in this specification are set forth in the solicitation. The date of the applicable DODISS and supplements thereto shall be as specified in the solicitation or contract. The issue in effect for drawings shall be as stated in the solicitation.

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation - Packaging, Methods of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-P-28809	Printed Wiring Assemblies
MIL-P-55110	Printed Wiring Assemblies
MIL-E-55585	Electronics Equipment and Parts, Packaging of

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A Notice 4	Electronic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-471	Maintainability Demonstration

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STANDARDS (Cont.)

MIL-STD-721B	Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors and Safety
MIL-STD-726	Packaging Requirements Code
MIL-STD-781B	Reliability Tests, Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1330	Definitions of Tactical, Prime Precise, and Utility of Terminologies for Classification of the DD Mobile Electric Power Engine Generator Set Family
MIL-STD-45662	Calibration Systems Requirements

DRAWINGS

B4027000	Generator, Digital Data SG-1139( )/G
SM-D-884567	CCA, Multi VDC
SM-D-884579	CCA, AC Input
B4027002	CCA, Transmit 1
B4027006	CCA, Transmit 2
B4027010	CCA, Receive 1
B4027014	CCA, Receive 2
B4027022	Cover Accessory Assembly
B4027042	Cover Assembly Front
B4027043	Control Filter Assembly
B4027047	Interface Box Assembly
B4027048	Cable Assembly Coaxial
SC-DL-525135	AC Power Cable Assembly, CX-11727
SM-D-571590-1	Adapters, BNC to Banana Plug
B4027049	Angle Assembly Mounting
B4027018	Case Assembly Electrical Equipment

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 FIRST ARTICLE (See 4.3). When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 CONSTRUCTION.

3.2.1 Fabrication. The equipment shall be constructed in accordance with DL-84027000 and the requirements specified therein.

3.2.2 Unit Serial Numbers. The SG-1139( )/G shall have consecutively assigned serial numbers unless otherwise specified by the contracting officer.

3.2.3 Printed Wiring. The printed wiring shall meet the requirements of MIL-P-55110 and MIL-P-28809.

3.2.4 Semiconductor Devices. The semiconductor devices shall meet the requirements of MIL-S-19500.

3.3 PARTS, MATERIALS, AND PROCESSES - GENERAL. Unless otherwise specified herein, parts, materials and processes shall be as shown on the drawings specified in 2. If a change or substitution is required, MIL-P-11268 shall apply. The paragraphs of MIL-P-11268 which are applicable to equipment of established design shall be followed. Submission of data shall be IAW applicable contract provisions.

3.4 FINISH. The equipment shall be finished in accordance with MIL-F-14072 and drawings specified in 3.2.

3.5 MARKING. The marking shall conform to MIL-M-13231, with approvals in accordance with requirements specified in the contract or purchase order.

3.5.1 Serial Numbers. Serial numbers shall be required on each nomenclatured item, except cords and cables.

3.6 PRECONDITIONING

3.6.1 Bounce Preconditioning. (See 4.6.1) Each assembled SG-1139( )/G shall meet the requirements specified herein without subsequent processing after submission to the bounce preconditioning of para. 4.6.1. Each SG-1139( )/G shall be subjected to this process.

3.6.2 Electrical Preconditioning. (See 4.6.2) Prior to submission to the specified quality conformance inspections, each assembled SG-1139( )/G consisting of the chassis, plug-in panel circuit card assemblies and control filter assemblies shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process.

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3.7 ELECTRICAL. (See 4.7)

3.7.1 General Characteristics. (See 4.7.1) Generator, Digital Data SG-1139( )/ shall produce and accept a pseudo-random data stream for the detection and display of errors induced during transmission.

3.7.2 Data Rates and Interfaces. (See 4.7.2)

3.7.2.1 Balanced Conditioned Diphase Modulated. The SG-1139( )/G shall provide and accept balanced conditioned diphase modulated data at rates of 0.6, 1.2, 2.4, 4.8, 9.6, 16 and 32 kilobits per second ±0.001 percent in accordance with the following interface criteria:

3.7.2.1.1 Driver Output Impedance. The driver output impedance shall be 100 ohms ±15%.

3.7.2.1.2 Driver Output Voltage. When loaded with a 100 ohm dummy load, the line-to-line voltage shall be 400 mv minimum and 4 volts maximum, peak-to-peak. When loaded with a 2400 ohm dummy load, center tapped to ground, the line-to-ground voltage magnitude shall not exceed 2 volts.

3.7.2.1.3 Driver Output Voltage Waveshape. When loaded with a 100 ohm dummy load, the rise and fall times (between 10% and 90% of the peak-to-peak voltage) shall be at least 0.78 usec and no more than 3.3 usec. There shall be no points of inflection prior to achieving 100% amplitude.

3.7.2.1.4 Input Voltage Range. A voltage between +50 mv and +3 volts shall cause the terminator to assume the mark state, and a voltage between -50 mv and -3 volts shall cause the terminator to assume the space state.

3.7.2.1.5 Operation. The terminator shall operate with line-to-ground signal peak voltage magnitude up to 3 volts, and line-to-ground DC voltage up to 56 volts.

3.7.2.1.6 Protection. Interface circuit protection shall be provided so that any driver or terminator shall not be damaged by occurrence of any of the following contingencies:

a. Voltage magnitude of less than or equal to 56 volts applied between any lead and ground.
b. Shorting of any leads of this interface to any other lead or to ground.
c. Opening of either or both of the input or output leads.
d. Application of a voltage transient through a surge impedance of 100 ohms between any lead and ground having the following wave shapes.

(1) A 1000 volt peak of either polarity, a rise to peak time of 5 microseconds, and a decay time to 50% of peak of 600 microseconds.

(2) A 350 volt peak of either polarity, a rise to peak time of 10 microseconds, and a decay time to 50% of peak of 2000 microseconds.

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3.7.2.1.7 Performance in the Presence of Noise. A driver connected to a terminator shall operate without error in the presence of longitudinal interference or a dc common return offset. With ±2 volts peak sinusoidal interfering signal at a frequency of 32 KHz and a DC offset voltage of ±56 volts present longitudinally, i.e., algebraically added to both terminator input terminals simultaneously with respect to the common return.

3.7.2.2 Unbalanced Non-Return to Zero. The SG-1139( )/G shall provide and accept unbalanced non-return to zero (NRZ) data at 576 kilobits per second ±0.001 percent with associated timing signals in accordance with the following interface criteria:

3.7.2.2.1 Impedance Requirements. The output impedance of drivers and the input impedance of terminators shall be 91 ohm ±10 percent.

3.7.2.2.2 Pulse Code Modulation (PCM OUT). The pulse width of the signal shall be 1.7 ±0.1 microseconds. The rise and fall times shall be 20 to 120 nanoseconds (ns). A digital "0" shall be a level between -2.2 and -2.6 volts. A digital "1" shall be a level between -0.4 V and +0.4 V. Ripple at either level shall be less than ±0.4 volts peak.

3.7.2.2.3 Timing (TIM OUT). The frequency of the signal shall be 576 KHz ±0.001 percent. The pulse width shall be 80 to 150 ns. The rise and fall times shall be 20 to 120 ns. The signal shall have a pulse peak 1.8 to 2.6 volts above a baseline of -2.0 to-2.3 volts. Ripple at the baseline shall be within ±.5 volts.

3.7.2.2.4 Phasing of PCM OUT and TIM OUT Signals. The leading edge of the TIM OUT pulses shall lead the transitions in the PCM OUT signal by the time ranges at the various temperatures as shown in the following phasing chart:

PHASING OF TIMING AND PCM SIGNALS

TEMPERATURE RANGE	MODE NUMBER OF CHANNELS	TIME RANGES (NS)
70° ±20°F and High Temperatures (+125°F)	12	-20 to -120
Low Temperature (-25°F)	12	0 to -100

3.7.2.3 Balanced Non-Return-to-Zero. The SG-1139( )/G shall provide and accept balanced non-return-to-zero (NRZ) data with associated square wave timing at rates of 128, 144, 256, 288, 512, 576, 1024, 1152, 1536, 2048, 2304, 4096 and 4608 Kb/s. Interface characteristics shall be as follows:

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3.7.2.3.1 Driver Output. The line-to-line voltage measured into a 100 ohm load center tapped to ground shall be between +100 mV and +4 volts for a digital one and between -100 mV and -4 volts for a digital zero. With the same load, the line-to-ground voltage magnitude shall be less than 3 volts.

3.7.2.3.2 Signal Balance. When measured as in 3.7.2.3.1, the difference in magnitudes between the one and zero voltages shall be such that the smaller is at least 90% of the larger.

3.7.2.3.3 Ringing. When measured as in 3.7.2.3.1, the voltage shall not overshoot its final value by more than 15%. Undershoot shall not reduce the signal to less than 100 mV.

3.7.2.3.4 Slew Rate. When tested as in 3.7.2.3.1, the transition between ±40 mV (either direction) shall take no more than 10 ns.

3.7.2.3.5 Timing Asymmetry. When measured as in 3.7.2.3.1, the positive half cycle of the timing wave shall have a width, measured at the zero voltage point, equal to half the bit interval ±15 ns.

3.7.2.3.6 Timing/Data Phasing. When measured as in 3.7.2.3.1, the negative to positive transition of the clock wave shall occur within ±29 ns of the data transitions. All times shall be measured at the zero voltage point.

3.7.2.3.7 Input Impedance. The terminator input impedance shall be 100 ohms ±10%, center tapped to ground.

3.7.2.3.8 Input Voltage Range. A voltage between -50 mV and +3 volts shall cause the terminator to assume the digital 1 state, and a voltage between -50 mV and -3 volts shall cause the terminator to assume the digital 0 state.

3.7.2.3.9 Operation. The terminator shall operate with line-to-ground voltage magnitude up to 3 volts.

3.7.2.3.10 Protection. The driver and receiver shall withstand without damage any of the following.

(1) Short, any line to ground

(2) Short, line-to-line

(3) Misconnection to another line carrying a signal covered by para. 3.7.2.3.1.

3.7.2.3.11 Additional Rates. When operating with the transmit section timed from the receive section or when timed from the external station clock, the SG-1139( )/G shall operate at any rate from 2 to 4915.2 Kb/s as determined by the external timing source.

3.7.3 Oscillator Frequency Accuracy. (See 4.7.3) The oscillator(s) generating the pseudo-random data streams shall have a frequency accuracy of ±0.001 percent when operating over the environmental temperature range specified (see 3.8).

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3.7.4 Functional. (See 4.7.4) In addition to the basic Data Generation/Error Detection capability, the SG-1139( )/G shall have the following features at required rates (see 3.7.2.1 for diphase rates, 3.7.2.2 for unbalanced, and 3.7.2.3 for balanced).

a. A capability for the display of:

(1) Errors
(2) Error rate

b. A capability for the injection, detection and display of single bit errors.

c. A capability to utilize an externally applied clock to generate the transmit data stream in accordance with 3.7.2.3.

3.7.5 Power Supply. (See 4.7.5) The power supply plug-in circuit card assemblies shall provide all DC voltages necessary for the operation of the internal circuits of the equipment. The input power shall not exceed 50 watts.

3.7.5.1 Primary power. The equipment shall operate from AC primary power having the following nominal ratings

a. 115 V, 50 Hz, single-phase.
b. 115 V, 60 Hz, single-phase.
c. 115 V, 400 Hz, single-phase.

The primary power shall have characteristics as contained in MIL-STD-1332 for Type 1, Class 1 and 2, Mode I, II, and III generator sets. In addition, the equipment shall be designed to operate from power having the following steady state voltage and frequency ranges

a. Voltage 115 VAC ±10 percent.

b. Frequency

1. 50 Hertz ±5 percent.
2. 60 Hertz ±5 percent.
2. 400 Hertz ±5 percent.

3.7.5.2 Internal power supply(s). The internal power supply(s) of the equipment shall provide the DC outputs for proper equipment operation which shall be kept to a minimum. The internal supply(s) shall be well regulated consistent with their function in the system. The internal supply(s) shall indicate no operational degradation in the performance over the entire range of the input power and environmental conditions specified herein.

3.7.5.3 Power supply monitor. When the power supply is properly energized, the power supply shall be in the "ON" state as indicated by an energized green indicator light located on the front panel of the equipment.

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3.7.6 Environmental Electrical Tests. (See 4.7.6)

3.7.6.1 Environmental functional test. At 32, 576 (ATACS), 4096 and 4608 Kb/s, the SG-1139( )/G unit under test shall be tested for the following features, using a Data Stream Generator (DSG) and an error inserter.

a. A capability for the display of

1. Errors
2. Error rate

b. A capability for the injection, detection and display of single bit errors.

c. A capability to run in the MASTER or LOOPED timing mode.

3.7.6.2 Environmental data rates and interface. For each of the following parameters, the indicated measurement shall be performed and compliance with the limi as specified in 3.7.2 shall be verified.

Diphase Data Out @ 32 Kb/s

- Upper and Lower Voltage Levels
- Rise and Fall times
- Wave shape

PCM Data Out, Tim Out @ 576 Kb/s

- Pulse width
- Upper and Lower Voltage Levels
- Rise and Fall times
- Phasing

Balanced NRZ @ 4096 and 4608 Kb/s

- Upper and Lower Voltage Levels
- Rise and Fall times
- Wave shape

3.7.6.3 Environmental oscillator accuracy

- Measure Timing Out @ 4608 and 4096 Kb/s.

3.7.7 Fault Location. An indicator shall be located on the front of each plug-in card for go-no go indication. The fault location circuitry shall activate the visual indicator on the board causing the fault. The equipment shall be capable of being maintained by standard military test equipment. Voltages and wave forms which are required in maintenance of the equipment shall be available at plug-in card connector pins.

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3.7.8 Accessory Items. The SG-1139( )/G shall be equipped with accessory items as follows:

a. One (1) each CX-11727 three-conductor AC power cable assembly in accordance with SC-DL-525135.

b. Four (4) each coaxial cable assemblies eight feet long (RG-62A/U with BNC connectors at each end).

c. One (1) each Interface Box consisting of aluminum chassis approximately 6" L X 4" W X 1.25" H that contain 16 banana jacks (8 in, 8 out) and two multi-pin connectors (MS3124E14-19P) wired as shown below. A chassis ground stud shall also be provided.

d. Four (4) each BNC to banana plug adapters.

MS3124E14-19P CONNECTOR	FUNCTION		
	BANANA JACK CONNECTORS		
PIN #	INPUT	OUTPUT	MISC
A	HIGH CHANNEL 1 IN		
B	LOW CHANNEL 1 IN		
C		HIGH CHANNEL 1 OUT	
D		LOW CHANNEL 1 OUT	
J	HIGH CHANNEL 2 IN		
K	LOW CHANNEL 2 IN		
L		HIGH CHANNEL 2 OUT	
M		LOW CHANNEL 2 OUT	
E	HIGH CHANNEL 3 IN		
F	LOW CHANNEL 3 IN		
G		HIGH CHANNEL 3 OUT	
H		LOW CHANNEL 3 OUT	
N	HIGH CHANNEL 4 IN		
P	LOW CHANNEL 4 IN		
R		HIGH CHANNEL 4 OUT	
S		LOW CHANNEL 4 OUT	
T	NC	NC	
U			CHASSIS GROUND
V	NC	NC	

3.8 SERVICE CONDITION TESTS. (See 4.8) The SG-1139( )/G shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted, or blistered; no moveable part or control shall be shifted in setting, position or adjustment; and no permanent degradation shall be caused in performance below that specified in the following environmental conditions.

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3.8.1 Temperature Test. (See 4.8.1)

3.8.1.1 Low temperature. (See 4.8.1.1) During and after testing as specified in 4.8.1.1, the equipment shall meet the requirements specified in 3.7.6.

3.8.1.2 High temperature. (See 4.8.1.2) During and after testing as specified in 4.8.1.2, the equipment shall meet the requirements specified in 3.7.6.

3.8.2 Humidity. (See 4.8.2) During and after testing as specified in 4.8.2, the equipment shall meet the requirements specified in 3.7.6. At the conclusion of this test, the equipment shall exhibit no physical damage such as corrosion, rust, blistering, swelling or deterioration of parts and materials. Any degradation in electrical or mechanical performance outside specification limits shall be considered a failure.

3.8.3 Low Pressure (Altitude). (See 4.8.3) During and after testing as specified in 4.8.3, the equipment shall meet the requirements specified in 3.7.6.

3.8.4 Vibration and Shock.

3.8.4.1 Vibration. (See 4.8.4.1) After testing as specified in paragraph 4.8.4.1, the equipment shall meet the requirements specified in paragraphs 3.7.6. and 3.7.6.3.

3.8.4.2 Shock (basic design). (See 4.8.4.2) After testing as specified in paragraph 4.8.4.2, the equipment shall meet the requirements specified in paragraphs 3.7.6.1 and 3.7.6.3.

3.8.4.3 Shock (bench handling). (See 4.8.4.3) After testing as specified in paragraph 4.8.4.3, the equipment shall meet the requirements specified in paragraphs 3.7.6.1 and 3.7.6.3.

3.8.4.4 Bounce (loose cargo). (See 4.8.4.4) After testing as specified in paragraph 4.8.4.4, the equipment shall meet the requirements specified in paragraphs 3.7.6.1 and 3.7.6.3.

3.8.5 Salt-fog. (See 4.8.5) After testing as specified in paragraph 4.8.5, the equipment shall meet the requirements specified in paragraphs 3.7.6.1 and 3.7.6.3, and when examined visually with the aid of a 10 power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surfaces, or exfoliation (See para 6.4.2) of metal.

3.8.6 Fungus. (See 4.8.6) After testing as specified in paragraph 4.8.6, the equipment shall be visually examined with the aid of a 10 power magnifier. The equipment shall exhibit no more than six minute unrelated spots, each no grater than 0.015 square inch in area, of sparse microbial growth (see para 6.4.4) as evidenced by growth colonization (see para 6.4.3) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see para 6.4.6) shall not be included in this evaluation.

3.8.7 Rain. (See 4.8.7) After testing as specified in paragraph 4.8.7, the equipment shall meet the requirements specified in paragraphs 3.7.6.1 and 3.7.6.3

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9 ELECTROMAGNETIC INTERFERENCE. (See 4.9) The SG-1139( )/G shall comply with e following requirements of MIL-STD-461A, Notice 4:

CE02	CS01	RE02 (a)	RS03 (b)
CE03	CS02	RE02.1	RS03.1 (b)
CE04 (a)	CS06		
CE05			

a. Conductec anc radiated switching transients shall comply with the quirements of MIL-STD-461A, Notice 4, Figure 5, and paragraph 6.13.4 spectively.

b. The electric field strength and modulation characteristic for radiated sceptibility shall be as follows.

FREQUENCY RANGE (Hz)	E-Field (V/M)	MODULATION	CHARACTERISTIC
10K - 2M	1	AM, 50%	1 KHz TONE
2M - 30M	5	AM, 50%	1 KHz TONE
30M - 76M	10	FM, 8 KHz Dev.	1 KHz TONE
76M - 400M	10	AM, 50%	1 KHz TONE
400M - 2G	10	PULSE 0.1 msec,	800 PPS
2G - 12.4G	5	PULSE 0.1 msec,	800 PPS

.9.1 Bonds and Grounds. The SG-1139( )/G shall be designec with adequate ovision for bonding the unit to the ground plane. Surface preparation for onc and ground sha'' be accomplishec by removing all high resistive elements rom the immediate area of contact. The bonding impedance between the two ating surfaces shall not exceed 2.5 milliohms. After ascertainment of the pecified bonding impedance and EMI tests, the prepared surfaces shall be estored with the original finish or other protective finish in accordance ith MIL-F-14072.

.9.2 Cables and Connectors. Coaxial cables, as required, shall be of the ouble-shielded type. Other cables requiring shields shall have a minimum of 0 dB attenuation to electric fields and plane waves over the frequency range f 10K - 10 GHz. All shielded cables shall have their shield terminated in an MI backshell to provide peripheral bonding of the shield. The connectors hall be free of all high impedance elements and provide positive bonding and rounding with mating connectors. The impedance between a single mating connection shall not exceed 2.5 milliohms.

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3.10 RELIABILITY. (See 4.10) The SG-1139( )/G shall have 2500 hour Mean-Time-Between Failures (MTBF) rate when operated in the normal modes of operation, and the following test conditions:

a. Temperature: 40 ±5°C

b. Temperature cycling: None.

c. Vibration: 2.2 G ±10% peak acceleration value at any nonresonant frequency between 20 Hz and 50 Hz, measured at the mounting points of the equipment. The duration of the vibration shall be ten (10) minutes of each operating hour.

d. Equipment on-off cycling: Equipment operating three (3) hours and fifty-five minutes and off five (5) minutes with this cycle continuing for the duration of the test.

e. Input voltage 115V AC ±5%

f. Input voltage cycling None.

3.11 INTERCHANGEABILITY. (See 4.11) Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454.

3.12 PANEL ASSEMBLIES (ELECTRICAL TEST). (See 4.12) All SG-1139( )/G circuit card assemblies and subassemblies shall meet the requirements of their respective Test Requirement Specification (TRS) drawings identified below.

Assembly No.	Name	TRS No.
B4027002	CCA, Transmit 1	B4027005
B4027006	CCA, Transmit 2	B4027009
B4027010	CCA, Receive 1	B4027013
B4027014	CCA, Receive 2	B4027017
B4027042	Control Filter Assembly	B4027045
SM-A-884567	CCA, Multi VDC	SM-A-884570
SM-A-884579	CCA, AC Input	SM-A-884582

3.13 SYSTEMS SAFETY ENGINEERING. (See 4.13)

3.13.1 Personnel Hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

3.13.2 Rounding Exposed Edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards, the design goal being a minimum radius of 0.04 inch (1 mm) for edges and a minimum of 0.5 inch (12.7 mm) for corners.

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3.13.3 Radioactive Materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/spark gaps, and lenses).

3.14 WORKMANSHIP. (See 4.13) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454, and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives
b. Cables, cords and wires, electrical
c. Wire (harnessing)
d. Plastic material and parts
e. Rubber
f. Wiring and cabling
g. Connectors (electrical)
h. Printed wiring assemblies

3.15 OPERATIONAL TEST. (See 4.7.1) Each fabricated and assembled SG-1139( )/G shall be subjected to a final Operational Test in accordance with 4.7.1 after all other inspections and tests are completed to determine that the SG-1139( )/G is operational prior to packaging. The Operational Test is a subset of the Functional Test of 4.7.4, conduct of a Functional Test satisfies this requirement.

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4. QUALITY ASSURANCE PROVISIONS

4.1 RESPONSIBILITY FOR INSPECTION. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Test equipment and facilities for government inspection shall be made available by the contractor.

4.2 CLASSIFICATION OF INSPECTION. Inspection shall be classified as follows:

a. First Article Inspection (does not include preparation for delivery (see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection:

1. Quality conformance inspection of equipment before preparation for delivery (see 4.5).

2. Quality conformance inspection of preparation for delivery (see 4.14

4.3 FIRST ARTICLE. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First Article Units. The contractor shall furnish 13 first article units of the complete SG-1139( )/G.

4.3.2 First Article Inspection. The first article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein except that all hardwired assemblies shall undergo in-process Group A inspection prior to bounce and electrical preconditioning.

4.3.3 First Article Data. The first article test plan and test report shall be as required in the contract or purchase order.

4.4 INSPECTION COVERED BY SUBSIDIARY DOCUMENTS. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

ITEM	WHERE REQUIRED
Parts, materials, and processes; general	3.3
Finish	3.4
Marking	3.5
Workmanship	3.14

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TABLE I

FIRST ARTICLE INSPECTION
(NOTES A & B)

INSPECTIONS	ROT PARA	PARA	UNIT 1	UNIT 2	UNIT 3	UNIT 4	UNIT 5	UNIT 6 - 13
1. INSPECTIONS COVERED BY SUBSIDIARY DOCUMENTS		4.4	INSPECTION TO BE PERFORMED ON ALL UNITS					
2. BOUNCE PRECONDITIONING	3.6.1	4.6.1	INSPECTION TO BE PERFORMED ON ALL UNITS					
3. ELECTRICAL PRECONDITIONING	3.6.2	4.6.2	INSPECTION TO BE PERFORMED ON ALL UNITS					
4. GROUP A INSPECTION	(SEE TABLE II)		INSPECTION TO BE PERFORMED ON ALL UNITS					
5. GROUP B INSPECTION	(SEE TABLE III)		INSPECTION TO BE PERFORMED ON ALL UNITS					
6. SAFETY	3.13	4.13	1					
7. EMI	3.9	4.9					1	
8. GROUP C INSPECTION								
LOW TEMPERATURE	3.8.1.1	4.8.1.1		2				
HIGH TEMPERATURE	3.8.1.2	4.8.1.2		3				
HUMIDITY	3.8.2	4.8.2			5			
LOW PRESSURE (ALT)	3.8.3	4.8.3		1				
VIBRATION	3.8.4.1	4.8.4.1			1			
SHOCK, BASIC DESIGN	3.8.4.2	4.8.4.2			2			
SHOCK, BENCH HANDLING	3.8.4.3	4.8.4.3			3			
BOUNCE LOOSE CARGO	3.8.4.4	4.8.4.4			4			
SALT FOG (NOTE C)	3.8.5	4.8.5		4				
FUNGUS (NOTE C)	3.8.6	4.8.6				1		
RAIN	3.8.7	4.8.7	2					
9. GROUP D INSPECTION								
RELIABILITY	3.10	4.10						1

NOTES A. The inspection 1 through 5, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

B. The numbers in the unit columns in the table covered by inspections 6, 7, 8 and 9 specify the order of inspections for the indicated unit.

C. The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

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4.5 QUALITY CONFORMANCE INSPECTION OF EQUIPMENT BEFORE PREPARATION FOR DELIVERY. The contractor shall perform the inspections specified in 4.5, 4.6 and 4.7. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A Inspection. Each unit (and subassembly and assembly identified in Table II) on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. The order of inspection of the subassemblies, assemblies and units (equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in Table II shall be subjected to a 100 percent visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in Table II.

b. After completion of the above 100 percent inspection by the contractor discrete lots shall be formed from the subassemblies and assemblies which passed the 100 percent inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQL's indicated in Table II (Quality Assurance verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. This requirement shall apply to spares only.

c. Assemblies from the discrete lots which have passed the above Quality Assurance verification inspection shall be assembled into the end equipment which shall then be subjected by the contractor to the preconditioning requirements of 4.6 and a 100 percent visual/mechanical and electrical inspection in accordance with Table II in that order.

NOTE Assemblies are not required to undergo further visual/mechanical and electrical inspection as part of the contractor's 100 percent Group A inspection of the equipment.

c. After completion of the above 100 percent inspection by the contractor, discrete lots shall be formed from units which passed the 100 percent inspection utilizing the procedures of MIL-STD-105 using the general inspection levels and AQL's indicated in Table II. Group A electrical inspection shall be performed in the order shown in Table II with the exception that the Funtional Test (4.7.4) shall be last.

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TABLE II

GROUP A INSPECTION

INSPECTION	ROT PARA	TEST PARA	AQL	
			MAJOR	MINOR
VISUAL/MECHANICAL (INCLUDING SPARE ASSEMBLIES)				
SG-1139( )/G CASE AND COVER	3.14	4.13	2.5*	10.0*
CONTROL FILTER ASSEMBLY	3.14	4.13	2.5*	4.0*
PANEL ASSEMBLIES	3.14	4.13	2.5*	4.0*
ELECTRICAL PERFORMANCE (INCLUDING SPARE ASSEMBLIES)				
CONTROL FILTER ASSEMBLY	3.12	4.12	1.0%	**
PANEL ASSEMBLIES	3.12	4.12	1.0%	**
ELECTRICAL FUNCTIONAL	3.7.4	4.7.4	1.0%	**

* Defects per hundred units.
** All electrical defects are major.

4.5.2 Group B Inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plan. The Group B AQL shall be as shown in Table III and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in the order listed in Table III except that interchangeability testing is done last.

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TABLE III

GROUP B INSPECTION

INSPECTION	ROT PARA	INSP PARA	AQL
INTERCHANGEABILITY	3.11	4.11	6.5%
DATA INTERFACES	3.7.2	4.7.2	4.0%
OSCILLATOR	3.7.3	4.7.3	4.0%
POWER VARIATION	3.7.5	4.7.5	4.0%

4.5.3 Group C Inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Subgroup 1. For this subgroup, one (1) unit shall be selected from the first production lot. For subsequent Group C inspection, one (1) unit shall be selected every third month. The unit shall be subjected to High and Low Temperature testing in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, two (2) units shall be selected from the first production lot. For subsequent Group C inspection, two (2) units shall be selected every fourth month. One (1) unit shall be subjected to EMI testing. One (1) unit shall be subjected to Vibration, Shock, Basic Design, Shock, Bench Handling, and Bounce, Loose Cargo testing in that order.

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TABLE IV

GROUP C INSPECTION

INSPECTION	ROT PARA	INSP PARA
SUBGROUP 1		
LOW TEMPERATURE	3.8.1.1	4.8.1.1
HIGH TEMPERATURE	3.8.1.2	4.8.1.2
SUBGROUP 2		
VIBRATION	3.8.4.1	4.8.4.1
SHOCK, BASIC DESIGN	3.8.4.2	4.8.4.2
SHOCK, BENCH HANDLING	3.8.4.3	4.8.4.3
BOUNCE, LOOSE CARGO	3.8.4.4	4.8.4.4
EMI	3.9	4.9
SUBGROUP 3		
HUMIDITY	3.8.2	4.8.2
LOW PRESSURE (ALTITUDE)	3.8.3	4.8.3
SALT FOG	3.8.5	4.8.5 (NOTE 1)
FUNGUS	3.8.6	4.8.6 (NOTE 1)
RAIN	3.8.7	4.8.7

NOTE 1 The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.5.3.1.2 Subgroup 2. For this subgroup, three (3) units shall be selected from the first production lot. For subsequent Group C inspection, three (3) units shall be selected every fifth month. One (1) unit shall be subjected to Humidity, Low Pressure (Altitude) and Rain testing in that order. One (1) unit shall be subjected to Salt Fog testing. One (1) unit shall be subjected to Fungus testing.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract.

4.5.4 Reinspection of Conforming Group C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

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4.5.5 Group D Inspection. This inspection shall consist of the tests specified in Table V and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection. Test conditions, methods and procedures shall be in accordance with paragraph 4.10.

TABLE V

GROUP D INSPECTION

INSPECTION	REQUIREMENT PARAGRAPH	INSPECTION PARAGRAPH
RELIABILITY	3.10	4.10

4.5.5.1 Sampling for Group D inspection. Eight (8) units shall be selected at random from the first month's production and subjected to the test of paragraph 4.10. For subsequent Group D inspections, eight units shall be selected at random from every sixth month's production.

4.5.5.2 Group D failures. Actions required relative to Group D failures shall be as specified in the contract.

4.5.6 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.6 Preconditioning.

4.6.1 Bounce Preconditioning. (See 3.6.1) Each unit shall be subjected to the following Bounce Preconditioning:

a. The unit, in its non-operating condition, shall be placed on the table of the package tester (Type 1000 SC as made by the L.A.B. Corporation, Skaneateles New York, or equivalent). The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for one (1) minute.

b. The unit shall then be inspected for visible evidence of damage, loose hardware and components and control positions. No control shall have changed position as a result of the test.

c. The unit shall then be subjected to electrical preconditioning.

4.6.2 Electrical Preconditioning. (See 3.6.2) Electrical preconditioning for the equipment and spare assemblies shall meet the requiremenets of 3.6.2.

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4.6.2.1 Test details.

a. The equipment shall be mounted on a vibration table in its normal operating position. Test Level B of MIL-STD-781B shall apply for Temperature and Vibration requirements. The equipment shall be connected as shown in Figure 1 or as specified by the specific test. The equipment shall be turned on and operated (all channels processing data) for 24 hours and then turned off for a maximum of one (1) hour followed by a second 24-hour operating period. At the end of the 48-hour preconditioning inspection, the equipment shall be subjected to the Operational Test per 4.7.1.

b. The chassis, plug-in printed circuit panel assemblies and control filter assembly shall be considered to have been successfully electrically preconditioned when they have individually passed a cumulative 48-hour operating preconditioning period without a failure during the last 24-hour period. If during the last 24-hour operating preconditioning period a plug-in panel assembly, control filter assembly or chassis fails, it may be replaced with a preconditioned panel, control filter assembly, or chassis and the preconditioning continued for a 24-hour failure-free period. The restart of a new 48-hour preconditioning period is not required. A plug-in panel assembly, control filter assembly or chassis failing operating preconditioning shall, after repair, be resubmitted to and pass the entire 48-hour operating preconditioning period prior to final electrical assembly tests by the contractor.

c. The unit chassis shall have no less than 48 hours of operational preconditioning nor more than 168 hours of operational preconditioning (exclusive of production test time) when used as a vehicle for preconditioning control filter assemblies and plug-in assemblies (including all spares).

d. Assemblies failing electrical acceptance tests subsequent to satisfactory completion of electrical preconditioning may be repaired with an electrically preconditioned assembly and electrically retested, accepted and delivered without repeat of electrical preconditioning.

4.6.2.2 Marking of electrically preconditioned units. All electrically preconditioned plug-in panel assemblies, control filter assemblies and chassis shall be physically identified by a preconditioning stamp. The stamp shall be affixed to the front panel near the nameplate of each unit which has successfully completed the preconditioning as outlined herein. The stamp on preconditioned plug-in panel assemblies shall be affixed to the bottom side, and to the bottom inside edge of the control filter assembly. The stamp shall be a "P" in a circle and shall be marked on the preconditioned unit per the Group III requirements of MIL-M-13231.

4.7 ELECTRICAL. (See 3.7) Unless otherwise specified all tests shall be performed at 115V ±5% from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure and humidity per MIL-STD-810 unless otherwise specified. The electrical test equipment listed in Table VI or their equivalent will be utilized during testing and will be calibrated in accordance with MIL-STD-45662. The standard test equipment shall be provided by the contractor. Government technical representative approval is required prior to use of proposed substitute items.

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4.7 ELECTRICAL (Cont)

Unless otherwise specified, special test equipment (See Table VII) shall be provided by the contractor subject to approval by the Government. Special test equipment is identified in the test figures, a brief description of checkout requirements and functions of this equipment is as follows

a. Data Stream Generator (DSG) described by DLB4027000 provides:

- Conditioned diphase data stream at 0.6, 1.2, 2.4, 4.8, 9.6, 16 and 32 kilobits per second.
- Unbalanced non-return-to-zero data stream at 576 kilobits per second with associated timing signals.
- Balanced non-return-to-zero data stream at 128, 144, 256, 288, 512, 576, 1024, 1152, 1536, 2048, 2304, 4096 and 4608 kilobits per second.
- An error detector which accepts any of the above data streams, correlates them with the transmitted data stream and provides an error count.

TABLE VI

STANDARD TEST EQUIPMENT

DESCRIPTION	MFG/MODEL NO.
Electronic Counter	H.P. 5425
DC Voltmeter	H.P. 412A
Signal Generator	H.P. 606B
RMS Voltmeter	H.P. 3403A
Audio Oscillator	H.P. 3325
Wattmeter	Weston 310
AC Voltmeter	H.P. 403
AC Power Source	Invertron 1501A
Oscilloscope	Tektronix 545
DC Power Supply	H.P. 6206B
Frequency Synthesizer	H.P. 3325

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4.7 ELECTRICAL (Cont.)

o Capability to inject single errors at any of the above data rates.

o Master or Looped Clock Signals at the above data rates.

b. Error Inserter provides the capability at any of the rates of (a) above to generate the following error rates in the data stream

Less than 10^{-5}
Greater than 10^{-5} and less than 10^{-3}
Greater than 10^{-3}

c. Data Rates and Interface Test Circuits provide the capability of adding voltages up to 56 volts to ground at the diphase leads and balanced non-return-to-zero leads and up to ± 2 volt sinusoidal interference signal.

TABLE VII

SPECIAL TEST EQUIPMENT

Equipment Name	Figure No.	Description
Data Stream Generator	5	Provides signals to SG-1139()/G under test and receives signals for error detection.
Error Inserter	5	See Schematic Fig. 5A
Data Rates and Interface Test Circuits	3A, 3B	See Schematic Fig. 3A & 3B

4.7.1 Operational Test. (See 3.7.1)

a. Connect equipment per Figure 1. Set timing selector switch to MASTER.

b. Connect AC power to the equipment under test (UUT). Turn on the AC power.

c. Operate data rate selector switch to 600 b/s (bits/seconds).

d. Reset readout utilizing front panel control.

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SG-1139( )/G

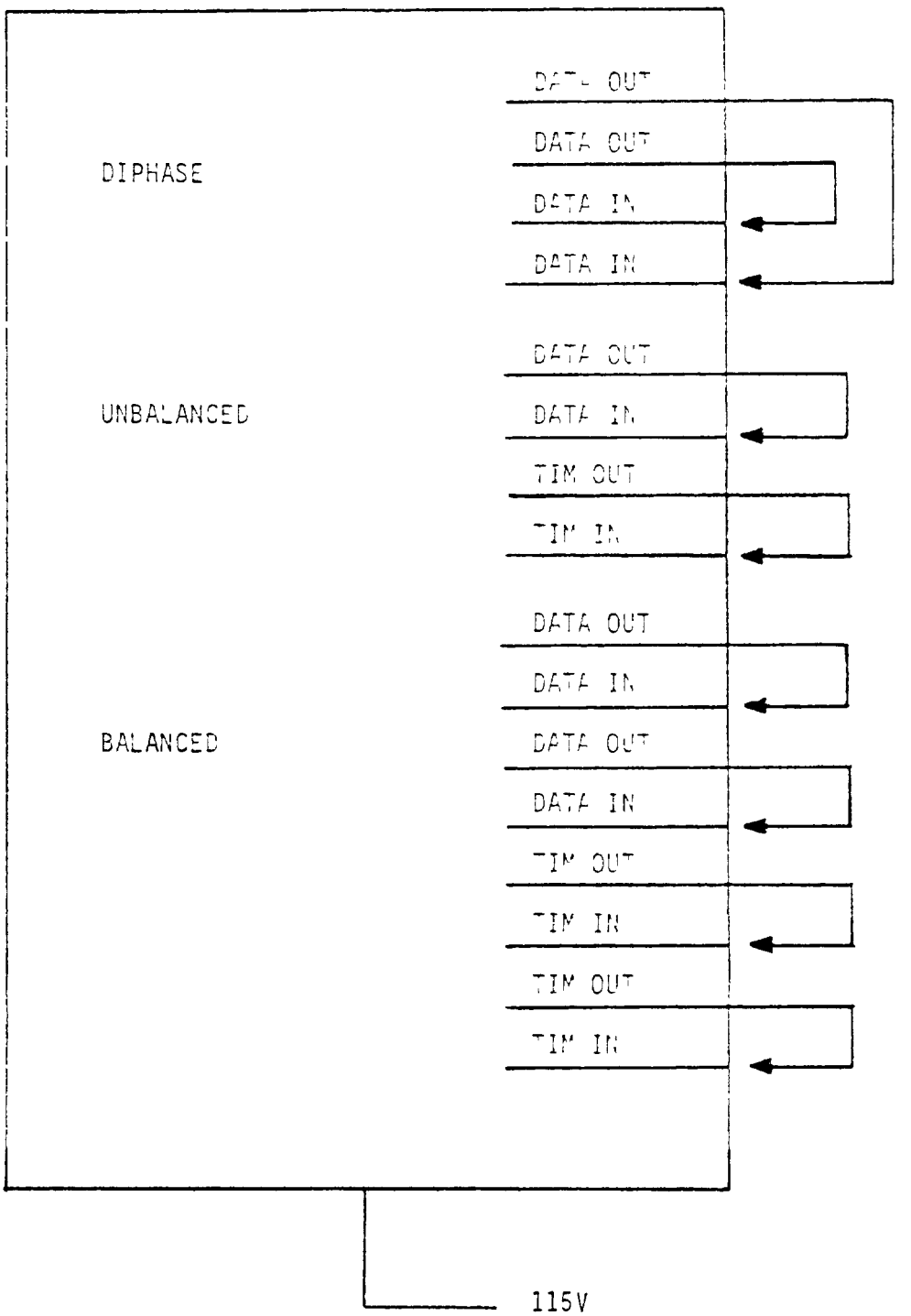


Figure 1. OPERATIONAL TEST

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4.7.1 Operational Test (Cont.)

e. Verify readout displays zero errors.

f. Inject one single bit error utilizing the front panel control.

g. Verify readout displays the single error injected.

h. Verify that only green error rate indicator is illuminated.

i. Reset the front panel display. Inject ten (10) single bit errors.

j. Verify that only red error rate indicator is illuminated.

k. Repeat steps c through h above for data rates 4096, 576 (ATACS) and 32 Kb/s.

l. With the data rate selector switch set to 32 Kb/s inject two (2) single bit errors.

m. Verify the yellow error rate indicator is illuminated.

4.7.2 Data Rates and Interface. (See 3.7.2)

a. Connect front panel interconnects per Figure 2.

b. Apply power to the equipment.

c. Operate data rate selector switch to 32 kilobits per second.

d. Measure the output waveform (Diphase DATA OUT) at the terminations for upper and lower voltage levels, rise and fall times, wave shape.

e. For the above parameters, the measurements shall meet the applicable requirements of paragraphs 3.7.2.1.1 thru 3.7.2.1.3.

f. Operate the data rate selector switch to 576 kilobits (ATACS) per second.

g. Measure the output waveform (DATA OUT, TIM OUT) at the terminations for pulse width, upper and lower voltage levels, rise and fall times and phasing. Using a counter, measure the frequency at the TIM OUT connector.

h. For the above parameters, the measurements shall meet the applicable requirements of paragraphs 3.7.2.2.1 thru 3.7.2.2.4.

i. Operate the data rate selector switch to 4608 and 4096 kilobits per second.

j. At each frequency, measure the output waveform (NRZ DATA OUT, TIMING OUT) at the terminations for driver output, slew time, timing asymmetry and phasing.

k. For the above parameters, the measurements shall meet the applicable requirements of paragraphs 3.7.2.3.1 thru 3.7.2.3.7.

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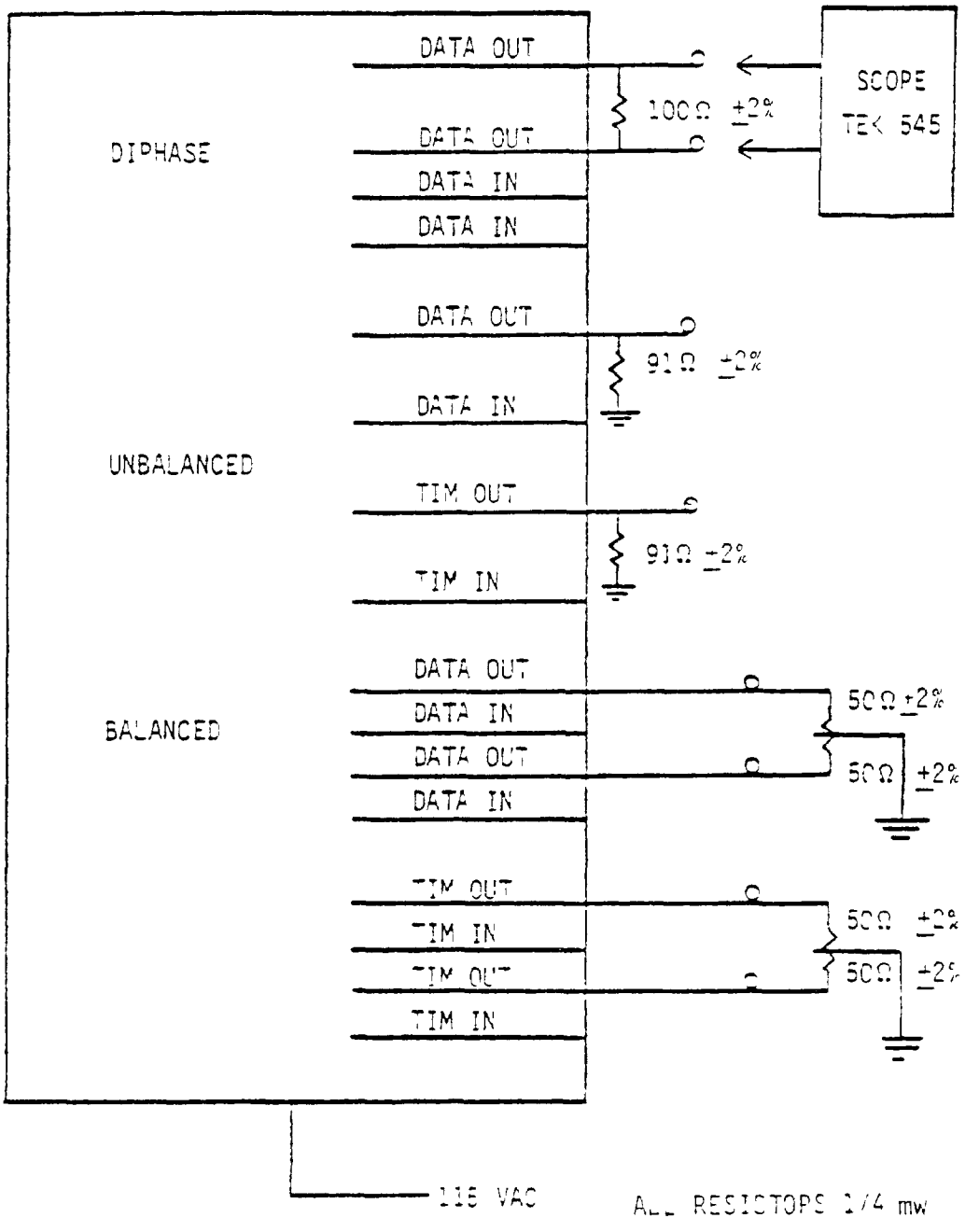


Figure 2. Data Rates and Interface Test

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4.7.2 Data Rates and Interface (Cont.)

l. Operate data rate selector switch to 32,000 bits per second.

m. Interconnect the circuit of Figure 3A between the Diphase DATA OUT and DATA IN terminals.

n. Adjust the DC power supply to 56 volts and the sine wave generator for a 4 volt peak-to-peak amplitude and a frequency of 32 KHz.

o. Reset readout utilizing front panel control.

p. Verify readout displays zero errors.

q. Inject one (1) single bit error utilizing front panel control.

r. Verify readout displays number of errors injected.

s. Repeat steps "o" through "r" above with the data rate selector switch set to 4608 Kb/s and the circuit of Figure 3B connected between the DATA OUT and DATA IN terminals. The DC power supply shall be adjusted so that the average voltage at the DATA IN terminals shall be -3 volts.

t. Operate data rate selector to 32,000 bits per second.

u. Interconnect a balanced attenuator between DATA OUT and DATA IN terminals.

v. Adjust the attenuator for a .100 volt peak-to-peak signal at the DATA IN terminal.

w. Inject single bit error(s) utilizing front panel control.

x. Verify readout displays the number of errors injected.

y. Repeat steps u through x as above with data rate selector switch set to 4608 kilobits per second.

4.7.3 Oscillator Accuracy Test. (See 3.7.3)

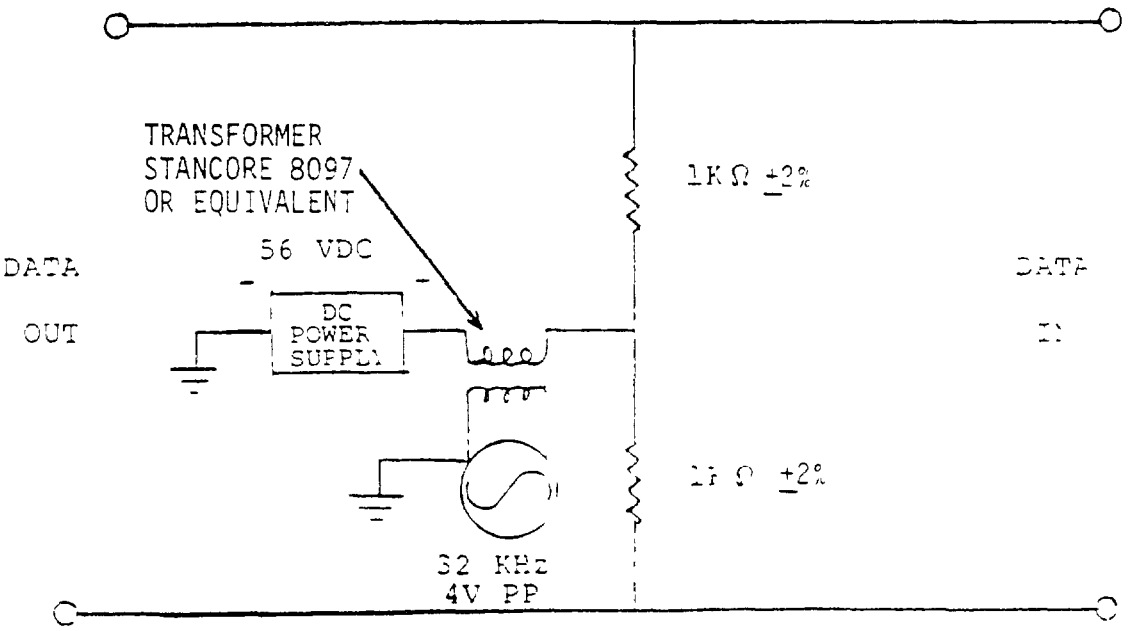
a. Connect equipment per Figure 4.

b. Apply power to the equipment.

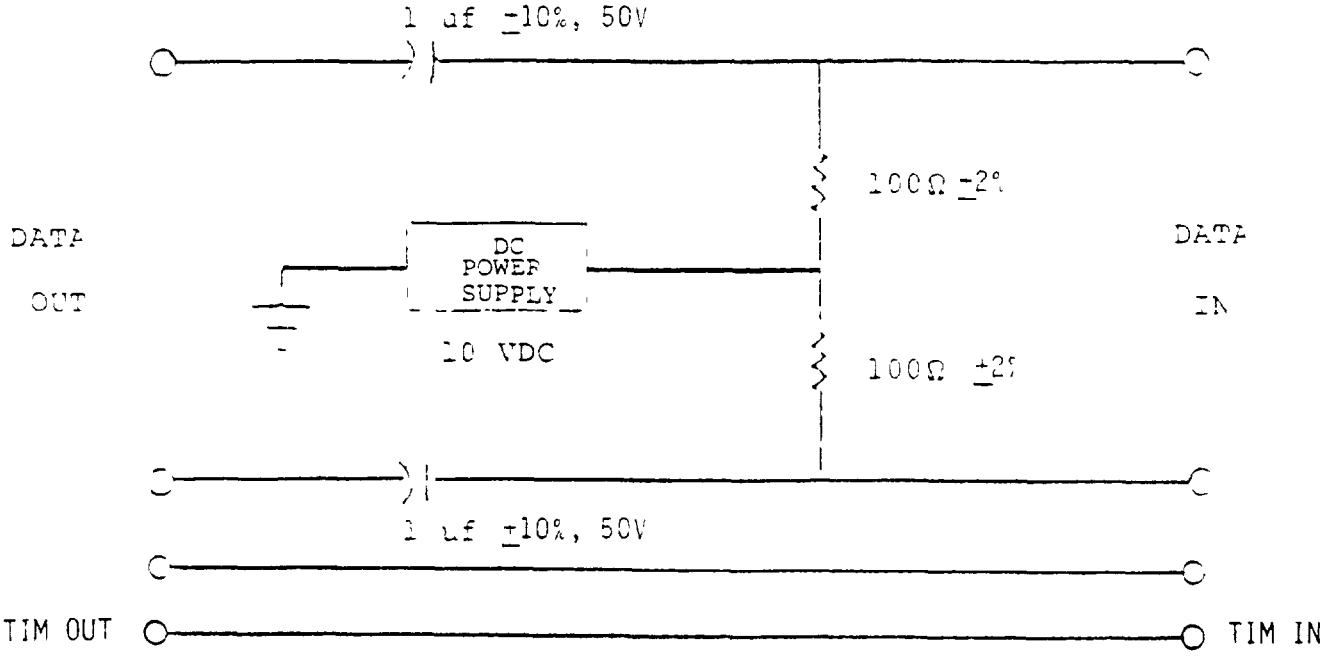
c. Measure the frequency at the Balanced NRZ TIM OUT connector for data rates 4096 Kb/s and 4608 Kb/s.

d. The measured frequency shall be within $\pm$0.001 percent of the data rate selected.

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DIPHASE
Figure 3A



BALANCED NRZ
Figure 3B

ALL RESISTORS 1/4W

DATA RATES AND INTERFACE TEST CIRCUITS

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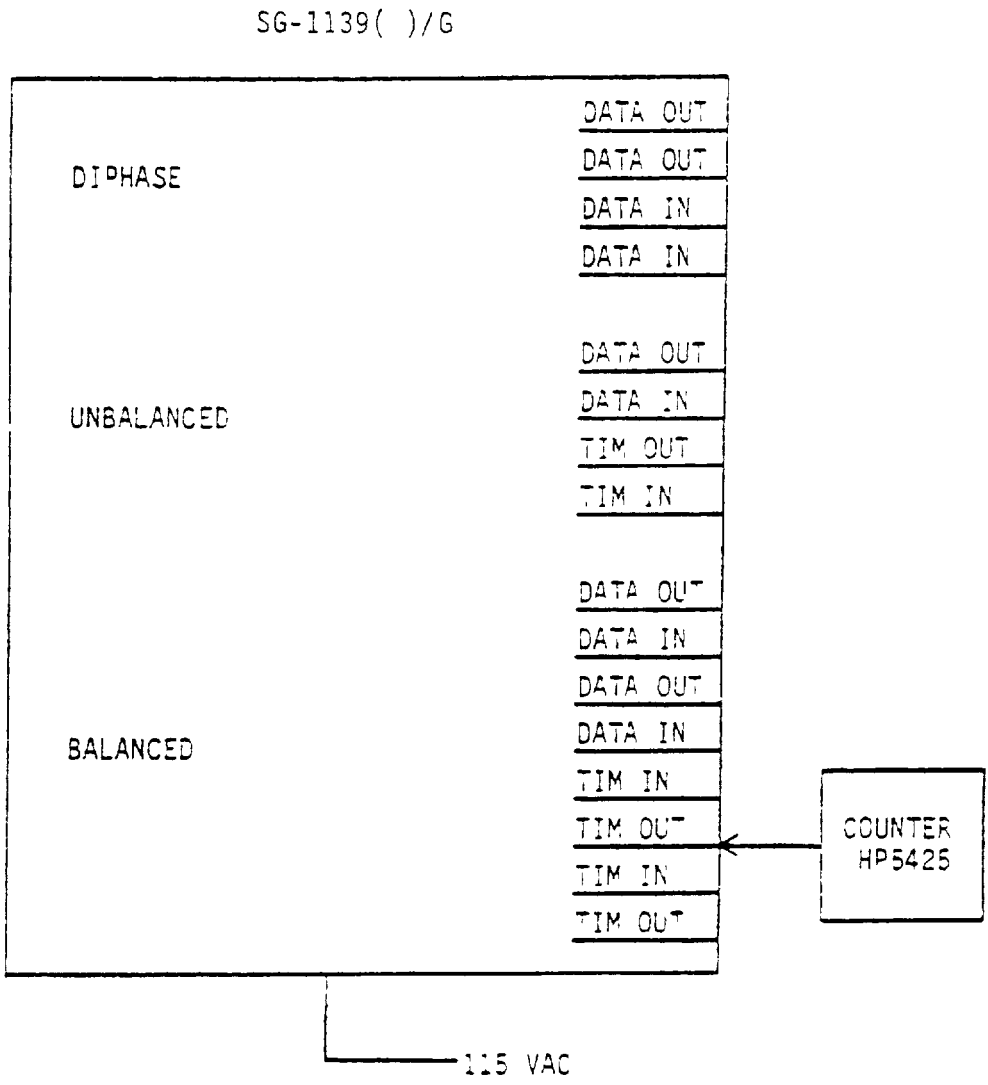


Figure 4 Oscillator Accuracy Test

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4.7.4 Functional Test. (See 3.7.4)

a. Connect equipment per Figure 5. The error inserter of Figure 5A shall be used in conjunction with the Data Stream Generator (DSG).

b. Apply power to the equipment. Set the timing selector switch on the DSG to the MASTER position and on the UUT SG-1139( )/G to the LOOPED position.

c. Set the data rate selector switch to 600 b/s on the UUT and DSG.

d. Reset the front panel displays and verify zero errors displayed.

e. On the DSG inject one single bit error.

f. Verify the UUT displays the injected error, and that the green error indicator is illuminated.

g. Repeat Steps c through f for all data rates. At rates 32, 576 (ATACS), 4096 and 4608 Kb/s, the following additional steps shall be performed.

1. Reset the UUT front panel display and set the error inserter to an error rate of greater than 10^{-5} and less than 10^{-3}. Verify UUT display is counting errors and the yellow error indicator is illuminated.

2. Reset the UUT front panel display and set the error inserter to an error rate of greater than or equal to 10^{-3}. Verify the UUT display is counting errors and the red error indicator is illuminated.

h. Set the timing selector switch on the UUT to the MASTER position. Set the timing selector switch on the DSG to the LOOPED position.

i. Set the data rate selector switch to 4608 Kb/s on both units.

j. Reset the front panel displays and on the UUT inject one single bit error

k. Verify the DSG unit display indicates the single error.

l. Repeat Steps i through k for data rates 4096, 576 (ATACS) and 32 Kb/s.

m. Connect the external clock to the STA CLK input on the UUT and set the UUT timing selector switch to the STA CLK position. Set the external clock to provide 4915.2 Kb/s.

n. Set the data rate selector switch on both units to 4608 Kb/s. Verify zero errors on both unit's display.

o. Inject one single bit error at the DSG unit and verify the UUT displays the single error and the green error rate indicator is illuminated.

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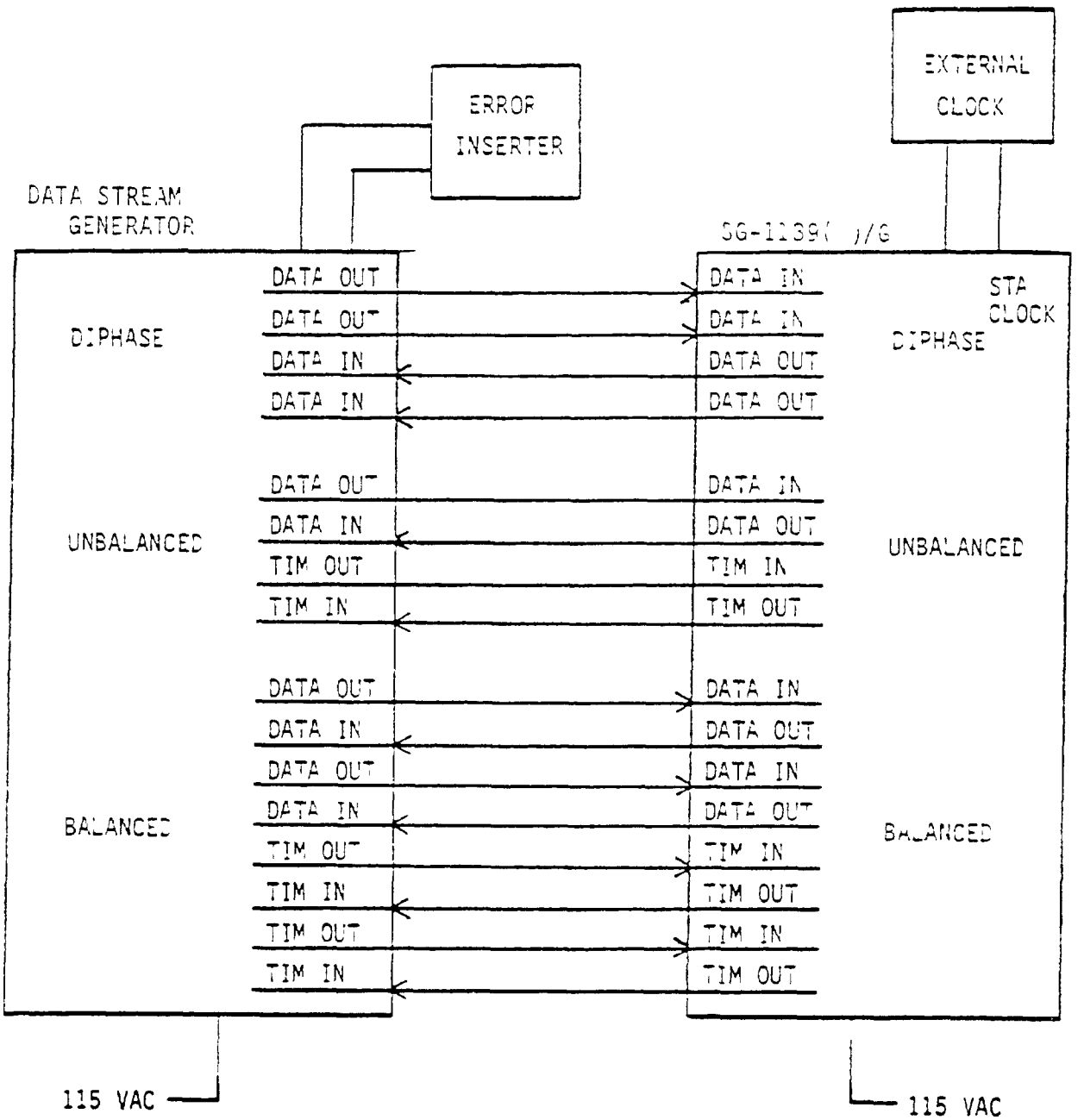


Figure 5. Functional Test

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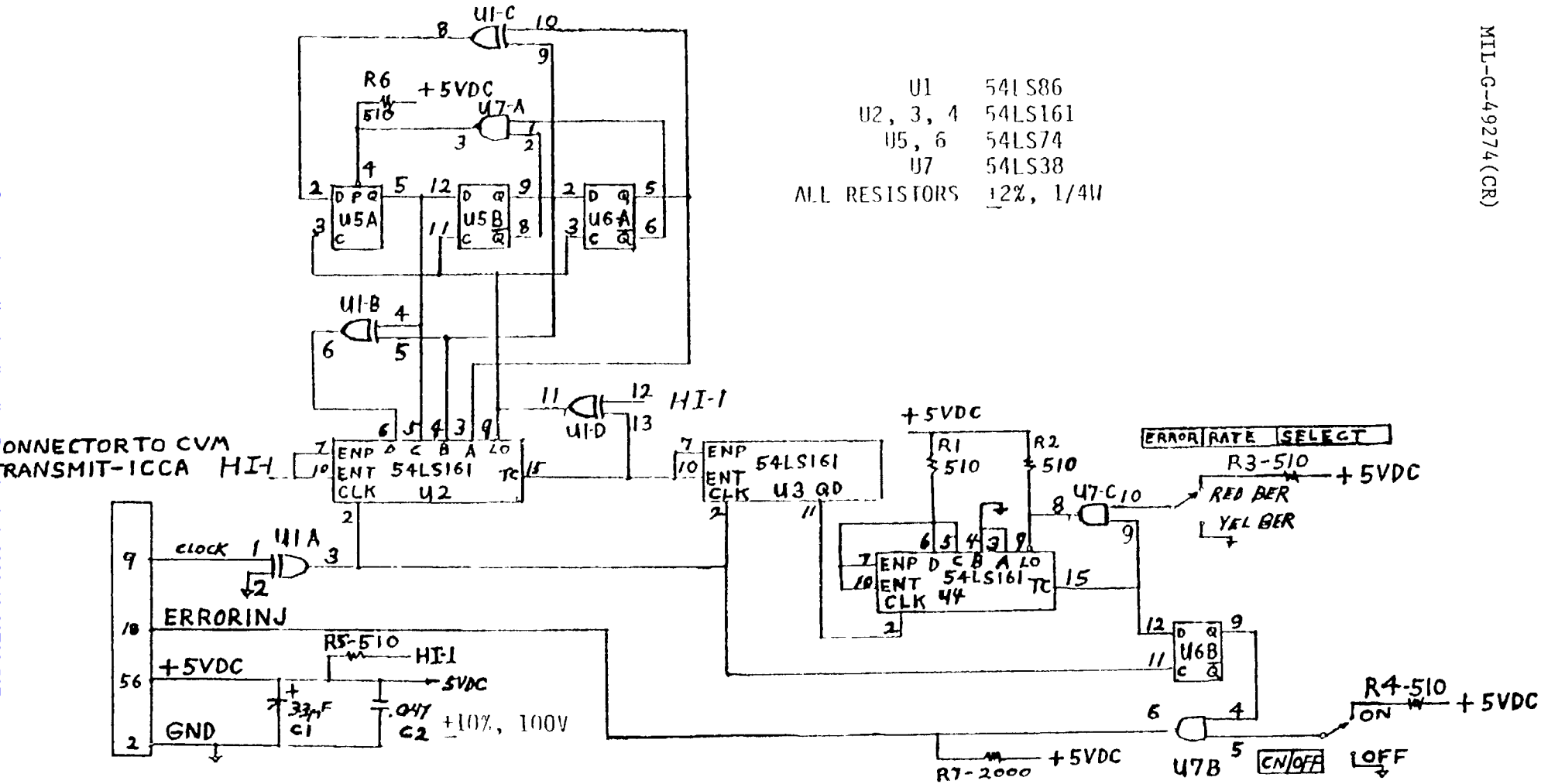


Figure 5A. Error Inserter Circuit

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4.7.4 Functional Test (Cont.).

p. At the DSG unit, set the error inserter to an error rate of greater than 10^{-5} and less than 10^{-3}.

q. Verify the UUT display is counting errors and the yellow error rate indicator is illuminated.

r. Set the error indicator to an error rate of greater than 10^{-3} and verify that the UUT display is counting errors and red error rate indicator is illuminated.

s. Set both units to 32 Kb/s. Disconnect the UUT Diphase Data Out from the DSG and connect it to a 100 ohm terminator. View the differential signal across the 100 ohms terminator with a Tektronix 545 scope or equivalent. Verify that the positive signal is at least 200 mv above ground and that the negative signal is at least 200 mv below ground.

4.7.5 Power Input Test. (See 3.7.5)

a. Connect equipment and the AC power line of the SG-1139( )/G per Figure 6.

b. Adjust the variable power supply for 60 Hz and 115 volts.

c. Operate the data rate selector switch to 4608 kilobits per second.

d. Reset readout using front panel control.

e. Inject single bit error(s) utilizing front panel control.

f. Verify readout displays number of errors injected.

g. Measure the AC power input to the SG-1139( )/G.

h. The input power shall not exceed 50 watts.

i. Perform the operational test per paragraph 4.7.1.

j. Repeat Steps g through i above with the power supply adjusted for the following AC line voltages and frequencies

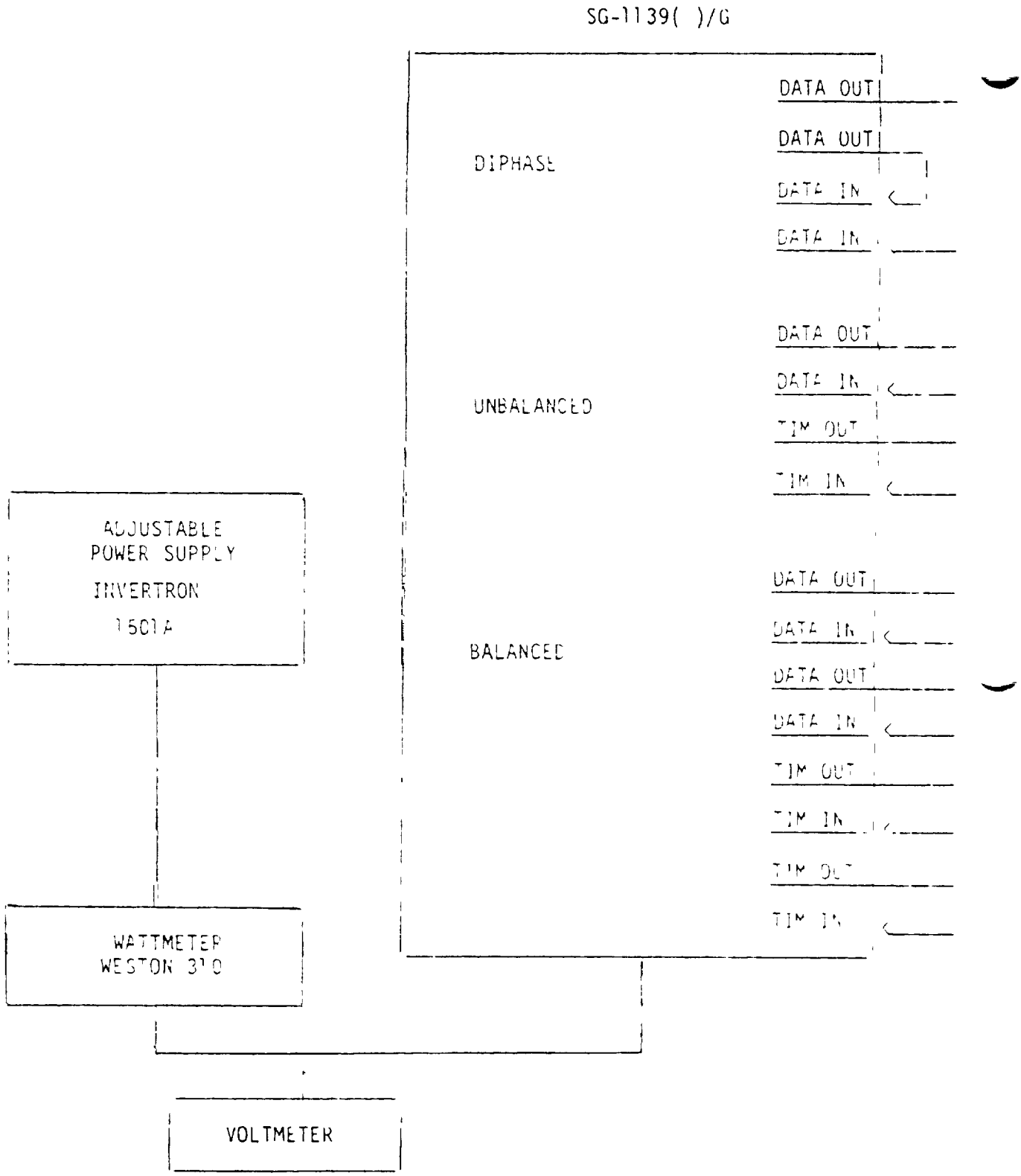
(1) 103.5 volts, 47.5 Hz

(2) 126.5 volts, 47.5 Hz

(3) 103.5 volts, 420 Hz

(4) 126.5 volts, 420 Hz

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POWER INPUT TEST

FIGURE 6

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4.7.6 Environmental Electrical Tests. (See 3.7.6)

4.7.6.1 Environmental Functional Test.

a. Connect equipment per Figure 5. The error inserter of Figure 5A shall be configured in the DSG.

b. Apply power to the equipment. Set the timing selector switch on the DSG unit to the MASTER position and on the UUT SG-1139( )/G to the LOOPED position.

c. Set the data rate selector switch to 32 Kb/s on the DSG and SG-1139( )/G units.

d. Reset the front panel displays and verify zero errors displayed.

e. On the DSG inject one single bit error.

f. Verify the UUT displays the injected error, and that the green error indicator is illuminated.

g. Reset the UUT front panel display and set the error inserter to an error rate of greater than 10^{-5} and less than 10^{-3}. Verify UUT display is counting errors and the yellow error indicator is illuminated.

h. Reset the UUT front panel display and set the error inserter to an error rate of greater than or equal to 10^{-3}. Verify the UUT display is counting errors and the red error indicator is illuminated.

i. Repeat Steps c through h for data rates 576 (ATACS), 4096 and 4608 Kb/s.

j. Set the timing selector switch on the UUT to the MASTER position and the DSG to LOOPED.

k. Set the data rate selector switch to 4608 Kb/s on both units.

l. Reset the front panel displays and on the UUT inject one single bit error.

m. Verify the DSG unit display indicates the single error.

n. Repeat Steps k through m for data rates 4096, 576 (ATACS) and 32 Kb/s.

4.7.6.2 Environmental data rates and interface test.

a. Connect front panel interconnects per Figure 2.

b. Apply power to the equipment.

c. Operate data rate selector switch to 32 kilobits per second.

d. Measure the output waveform (Diphase DATA OUT) at the terminations for upper and lower voltage levels, rise and fall times, wave shape.

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4.7.6.2 Environmental data rates and interface test (Cont.)

e. The appropriate requirements of paragraph 3.7.2.1 shall be met.

f. Operate the data rate selector switch to 576 (ATACS) kilobits per second.

g. Measure the output waveform (DATA OUT, TIM OUT) at the terminations for pulse width, upper and lower voltage levels, rise and fall times and phasing.

h. The appropriate requirements of paragraph 3.7.2.2 shall be met.

i. Operate the data rate selector switch to 4608 and 4096 kilobits per secon

j. Measure the output waveform (NRZ DATA OUT, TIMING OUT) at the termination for driver output, slew time, and timing asymmetry.

k. The appropriate requirements of paragraph 3.7.2.3 shall be met.

4.7.6.3 Environmental oscillator accuracy test.

a. Connect equipment per Figure 4.

b. Apply power to the equipment.

c. Measure the frequency at the balanced NRZ TIM OUT connector for data rates 4096 Kb/s and 4608 Kb/s.

d. The measured frequency shall be within ±.001 percent of the data rate selected.

4.8 SERVICE CONDITION TESTS. (See 3.8)

4.8.1 Temperature Tests.

4.8.1.1 Low temperature. (See 3.8.1.1) The equipment shall be tested in accordance with Test Method 502.1, Procedure I of MIL-STD-810. Storage temperature shall be minus 70 degrees F. Storage time shall be 24 hours. Operating temperature shall be minus 25 degrees F. At operating temperature, the test shall start within 5 minutes after power application to the unit. Performance tests, as specified in 4.7.6 shall be made during and after testing, as specified in Test Method 502.1.

4.8.1.2 High temperature. (See 3.8.1.2) The equipment shall be tested in accordance with Test Method 501.1, Procedure II of MIL-STD-810. Operating temperature shall be plus 125 degrees F. Performance tests, as specified in 4.7.6 shall be made during and after testing, as specified in Test Method 501.1.

4.8.2 Humidity Test. (See 3.8.2) The equipment with all covers removed, shall be tested in accordance with Test Method 507.1, Procedure II of MIL-STD-810. Performance testing as specified in 4.7.6.1 shall be made during each 24 to 28 hour test period. During the last 5 hour period of the last 48-hour cycle conduct testing as specified in 4.7.6. At the conclusion of testing, performance

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4.8.2 Humidity Test (Cont.)

testing as specified in 4.7.6 shall be made. Failure of the equipment to meet the requirements of 3.8.2 shall constitute failure of this test.

4.8.3 Low Pressure Altitude Test. (See 3.8.3) The equipment shall be tested in accordance with Test Method 500.1, Procedure I of MIL-STD-810. In Step 2, the rate of altitude change shall not exceed 2500 -250 ft. per minute. Performance tests as specified in 4.7.6.1 shall be made during testing as specified in Test Method 500.1. After testing, performance tests, as specified in 4.7.6 shall be made.

4.8.4 Vibration and Shock. (See 3.8.4)

4.8.4.1 Vibration test. (See 3.8.4.1) The equipment shall be tested in accordance with Test Method 514.2, Procedure VIII of MIL-STD-810. Table 514.2-VI for equipment condition "wheeled vehicles" and Curve V of Figure 514.2-6 shall be used. Performance testing as specified in 4.7.6.1 and 4.7.6.3, shall be made after testing as specified in Test Method 514.2.

4.8.4.2 Shock (basic design) test. (See 3.8.4.2) The equipment shall be tested in accordance with Test Method 516.2, Procedure I of MIL-STD-810. Performance testing as specified in 4.7.6.1 and 4.7.6.3 shall be made after testing as specified in Test Method 516.2, Figure 516.2-2 (peak value (A) g's-20, nominal duration (D)-11 ms).

4.8.4.3 Shock (bench handling) test. (See 3.8.4.3) The equipment shall be tested in accordance with Test Method 516.2, Procedure V of MIL-STD-810. Performance testing as specified in 4.7.6.1 and 4.7.6.3 shall be made after testing as specified in Test Method 516.2.

4.8.4.4 Bounce, Loose Cargo. (See 3.8.4.4) The equipment shall be tested in accordance with Method 514.2, Procedure XI, Part 2 of MIL-STD-810. Performance testing as specified in 4.7.6.1 and 4.7.6.3 shall be made after testing as specified in Method 514.1.

4.8.5 Salt Fog Test. (See 3.8.5) The equipment shall be tested in accordance with Test Method 509.1 of MIL-STD-810. At the conclusion of the test, the equipment shall be examined for evidence of visible corrosion. Performance testing as specified in 4.7.6.1 and 4.7.6.3 shall be made after testing as specified in Test Method 509.1. Failure of the equipment to meet the requirements of 3.8.5 shall constitute failure of this test.

4.8.6 Fungus Test. (See 3.8.6) The equipment, with all covers removed, shall be tested in accordance with Test Method 508.1, Procedure I of MIL-STD-810. There shall be abundant growth colonization (see 6.4.3) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of equipment together with cases and accessories, is permitted for 72 hours prior to the start of fungus test. Failure of the equipment to meet the requirements of 3.8.6 shall constitute failure of this test.

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4.8.7 Rain Test. (See 3.8.7) The equipment in its transit configuration shall be tested in accordance with Test Method 506.1, Procedure II of MIL-STD-810. After testing, performance tests as specified in 4.7.6.1 and 4.7.6.3 shall be made

4.9 ELECTROMAGNETIC INTERFERENCE TESTS. (See 3.9) The SG-1139( )/G shall be tested for compliance with the requirements of 3.9. Test procedures shall be in accordance with the test methods of MIL-STD-462, Notice 3. A test plan must be prepared by the contractor and submitted to the Government for approval prior to the start of EMI testing. Prior to performing the EMI tests, the bonding requirements of 3.9.1 and 3.9.2 shall be ascertained. The bonding impedance shall be measured and the data recorded in the EMI test report.

4.10 RELIABILITY. (See 3.10)

4.10.1 First Article Reliability Test. The first article reliability inspection testing shall be in accordance with Test Plan IV of MIL-STD-781B.

4.10.2 Quality Conformance (Group D) Reliability Test. The reliability test for Group D inspection shall be in accordance with Test Plan IVa of MIL-STD-781B.

4.10.3 Reliability Test Procedure. The detail test procedures and methods for the tests described in 4.10.1 and 4.10.2 shall be in accordance with contractor prepared, Government approved Test Plan.

4.10.4 Reliability Test Conditions.

4.10.4.1 Performance parameters to be measured. Performance testing as specified in paragraph 4.7.1 shall be performed on all units subjected to reliability testing. Performance testing to requirements of 4.7.1, 4.7.2 and 4.7.3 is to be completed before and after first article reliability testing only.

4.10.4.2 Environmental and operational stresses. Test Level B of MIL-STD-781B shall apply except as noted below

a. Input voltage cycling specified in Test Level B does not apply.

b. Equipment ON-OFF cycling is revised to read "Equipment ON-OFF Cycling. The equipment shall be turned on and operated with all channels processing data for 3 hours, 55 minutes, then turned off for 5 minutes. This cycle shall continue throughout the test."

c. The Operational Test of 4.7.1 shall be performed during the "ON" period of every sixth cycle. The time period expended in the performance of the Operational Test shall not exceed the "ON" time. Performance of the test shall be limited to normal workdays (excluding weekends).

4.10.5 Vibration. During testing, the equipment shall be mounted on a vibration table positioned in its normal operating mode with external covers removed. Vibration shall be in a plane perpendicular to the printed circuit cards.

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4.10.6 Failure Reporting and Analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781B. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.10.7 Failure Definition

a. Definitions. The definitions contained in MIL-STD-471, MIL-STD-781B and MIL-STD-721B will apply except as modified herein.

b. Failure. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below performance limits specified in 4.7.1.

c. Failure Categories. Failures will be categorized as relevant or non-relevant failures.

d. Relevant Failures. A relevant failure is one that can be attributed after failure analysis to any of the following causes

(1) Design defects.

(2) Manufacturing errors.

(3) Workmanship defects.

(4) Physical deterioration (such as wearout or fatigue).

(5) Tolerance degradation beyond the limits defined on the data sheets.

(6) Failure of alarm or test circuitry or indicator lamps to indicate a failure or that affect system operation.

Failures that are not proven to be non-relevant will be considered as relevant.

e. Non-Relevant Failures.

(1) Damage resulting from improper installation.

(2) Failure of test instrumentation or monitoring equipment which is external to the unit on test.

(3) Damage resulting from accident or mishandling.

(4) Failure due to technician or operator errors.

(5) False alarms caused by test circuitry failure.

(6) Dependent failures as defined by MIL-STD-781B except that paragraph 5.5.1(2) of MIL-STD-781B shall govern in the event of multiple failures.

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4.10.8 Accept-Reject Criteria. The accept-reject criteria shall be Test Plan IV or IVa of MIL-STD-781B.

4.11 INTERCHANGEABILITY INSPECTION. (See 3.11) Interchangeability shall be proven by disassembling all normally removable modules, boards and parts from Group B sample units that have been subjected to and met all other Group B tests, interchanging approximately one-half of the parts (while assuring that one pair of each type part is interchanged) and reassembling. The units shall meet the requirements of 3.11 and pass Operational Test of 4.7.1.

4.12 ELECTRICAL SUBASSEMBLY TESTS. (See 3.12) Plug-in subassemblies, including spares, shall meet the requirements of their respective test requirements specification drawings listed in 3.12.

4.13 VISUAL AND MECHANICAL INSPECTION. (See 3.13 and 3.14) Each device and accessory shall be visually and mechanically examined for the defects listed in MIL-STD-252 and those features specified in paragraph 3.13 and 3.14 herein. Parts, components and complete equipments shall be inspected for workmanship, mechanical construction, loose nuts and screws, application of specified treatments, safety (see 3.13) and miscellaneous defects. The equipment and controls shall be inspected for proper mechanical and electrical operation. Wiring, soldered connections, ground welds, finishes, etc. shall be inspected for workmanship. Clearance, dimensions and mechanical adjustments shall be measured. Labels and markings shall be inspected. Examination shall also be made for other visual or mechanical defects, similar to those described above that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government as major or minor in accordance with the definitions of MIL-STD-105 and MIL-STD-252.

An inspection shall be performed to verify compliance with those portions of 3.13 (System Safety Engineering) which can be determined visually.

4.14 QUALITY CONFORMANCE INSPECTION OF PACKAGING. The packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PACKAGING.

5.1 PRESERVATION. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. JR-1-1-ØØ-00-NS-X-ED-Ø-00-A.

5.1.2 Level B. JR-1-1-ØØ-00-NS-X-ED-Ø-00-B.

5.2 PACKAGING AND MARKING. Packaging and marking shall be in accordance with MIL-E-55585 and conform to the applicable clause of the contract or purchase order.

6. NOTES.

6.1 ORDERING DATA. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

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6.1 ORDERING DATA (Cont.)

b. Level A or B preservation and packing. (See Section 5)

c. Marking and shipping of samples.

d. Place of final inspection.

e. Technical literature required, test tools and running spares.

f. First article inspection. (See 4.3)

g. Length of cables (if any are required).

6.2 ENVIRONMENTAL. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.3 GROUP C INSPECTION. (See 4.5.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the Contracting Officer on the adequacy of corrective action.

6.4 DEFINITIONS.

6.4.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.4.2 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.4.3 Growth Colonization. Growth Colonization is a mass of individual plants, generally of one species, living together or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.4.4 Microbial Growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.4.5 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.4.6 Tubular Germination. Tubular germination is partial growth by the reproductive of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

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6.5 LOCATION OF OPERATIONAL INSPECTION. (See 3.15) It is desirable that the Operational Test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 VERIFICATION INSPECTION. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established by the Government and will be adjusted to make maximum use of the contractor's quality system and the quality history of the product.

CUSTODIAN:

Army-CR

Preparing Activity

Army-CR
(Project Number 6625-A821)

User Activities:

Navy - MC

Air Force - 13

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-G-49274(CR)	Generator, Digital Data SG-1139( )/G

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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